

Description

A 38 Year-old adult patient is seen by her Primary Care Provider at NIST Primary Care on March 15, 2025 at 2pm for pain and swelling in the arm following a fall. A clerical assistant registers the patient. She records the patient's name, date of birth, race, ethnicity, residence, insurance information (PlanID123 from MyInsuranceCompany), and health history. The clerical assistant also records the patient's chief complaint in free-text as, "Pain and swelling in the forearm following a fall."

NIST Primary Care is an outpatient facility that routinely sends electronic syndromic surveillance data to the State Public Health Department (NIST PH SS FAC 2) syndromic surveillance program in accordance with a state regulation. At 2:30 PM, the facility's electronic health record module for syndromic surveillance data assembles and transmits a Registration ADT message about this patient encounter. The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions.

Comments

Dates and times are provided in this test scenario to illustrate the sequence of clinical and messaging events. Since the exact dates and times are not reproducible when modeling the test scenario with HIT, only date and time format will be validated within tester submitted test data.

The dates within the message and across the messages in this test case will be subject to visual inspection to verify data consistencies and the ability to conform to the requirement to send this message within 12 hours of the event triggering this message.

Pre-condition

No Pre-Condition

Post-Condition

An ADT^A04 Registration message has been triggered.

Test Objectives

This test step examines an HIT Module's ability to create an ADT^A04 Registration message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance requirements.

- Setting is Ambulatory.
- Chief complaint is captured as narrative in PV2-3 Admit Reason
- **Patient Name is Identifiable in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in an identifiable manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:** Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Pediatric case where age is measured in months.
- Multiple Races
- Insurance known

Evaluation Criteria

Message Validation Report

Diagnosis code (DG1-3) validation is validated as follows:

The table below summarizes the format representations that are accepted for ONC Health IT Module Certification. Rows 1 and 2 show the formats as required by the implementation guide and standard terminology rules and best practices. The last 3 columns show acceptable alternative formats. For this test case one of the following values shall be present in the message:

DG1(1)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
M84332A	Stress fracture, left ulna, initial encounter for fracture	I10C	M84332A^Stress fracture, left ulna, initial encounter for fracture^I10C	Yes
11313841000119108	Stress fracture of left ulna (disorder)	SCT	14189004^Stress fracture of left ulna (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
M84.332A	Stress fracture, left ulna, initial encounter for fracture	I10C	M84.332A^Stress fracture, left ulna, initial encounter for fracture^I10C	Yes
M84332A	Stress fracture, left ulna, initial encounter for fracture	I10	M84332A^Stress fracture, left ulna, initial encounter for fracture^I10	No
M84.332A	Stress fracture, left ulna, initial encounter for fracture	I10	M84.332A^Stress fracture, left ulna, initial encounter for fracture^I10	No
54556006	Fracture of ulna (disorder)	SCT	74325003^Fracture of ulna (disorder)^SCT	Yes

The NIST column indicates what the NIST Tool Validation will accept in context-based testing (used for ONC Certification). A 'Yes' in the NIST column signifies that the tool will accept the format and not throw an error. A 'No' in the NIST column signifies that the tool will throw an error (specifically for DG1-3.3). However, based on the recommendation from the CDC SMEs and approval from ASTP/ONC the format is acceptable. The NIST test cases documents the instances where the test proctor can grant an exception when the tool issues and error.

Notes

Messages in this test case will be used in a Manual Test Step under functional requirement testing.